

Exo: 01

1/ La population : Les femmes souffrant d'anémie. $N=20$.

2/ Le type des caractères statistiques :

groupe sanguin $\rightarrow$ caractère
Age $\rightarrow$ quantitatif continu

Nombre d'enfants $\rightarrow$ " discret.

3/

x_i	n_i	n_i N_i	$n_i \cdot x_i$	$n_i - x_i^2$
0	4	4 0	0	0
1	6	6 4	6	6
2	5	10 10	10	20
3	2	15 12	6	18
4	2	17 14	8	32
5	0	19 14	0	0
6	1	19	6	36
Σ	20	20	36	112

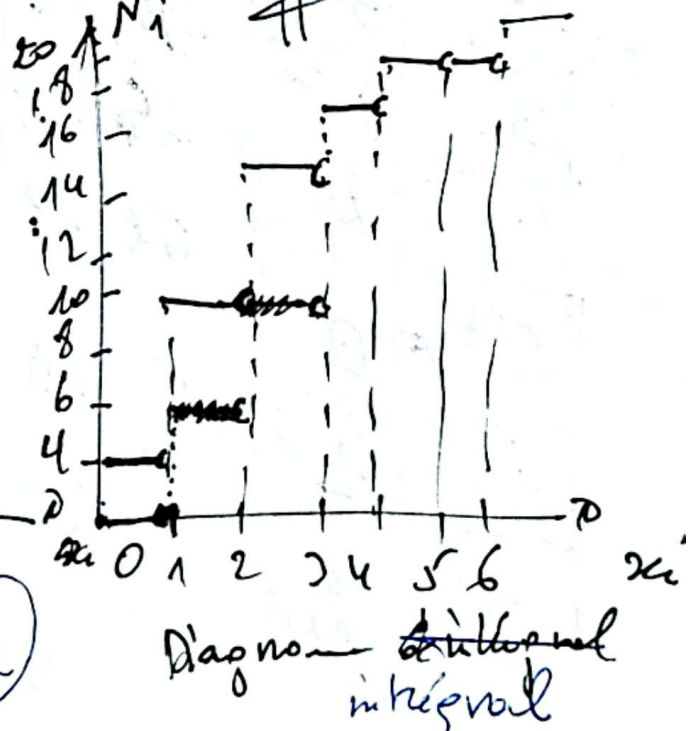
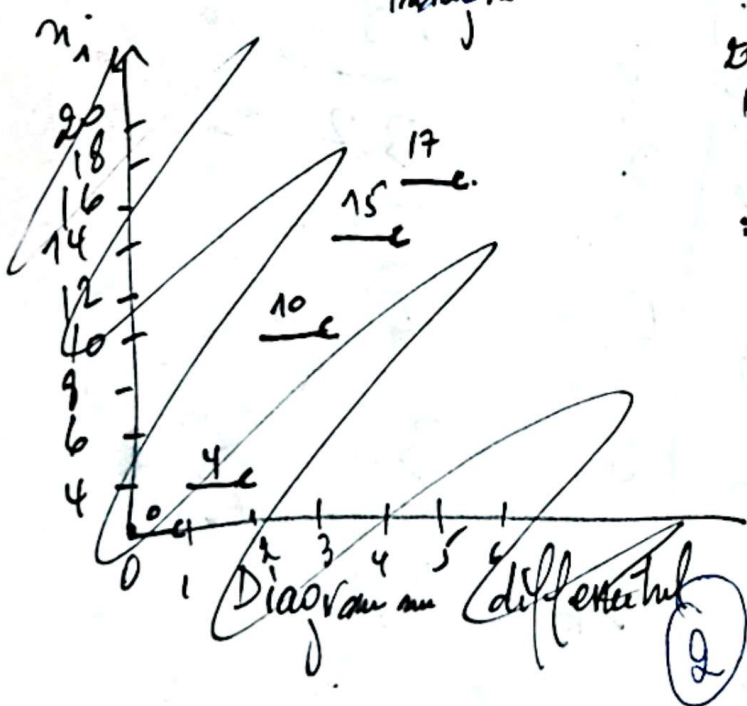
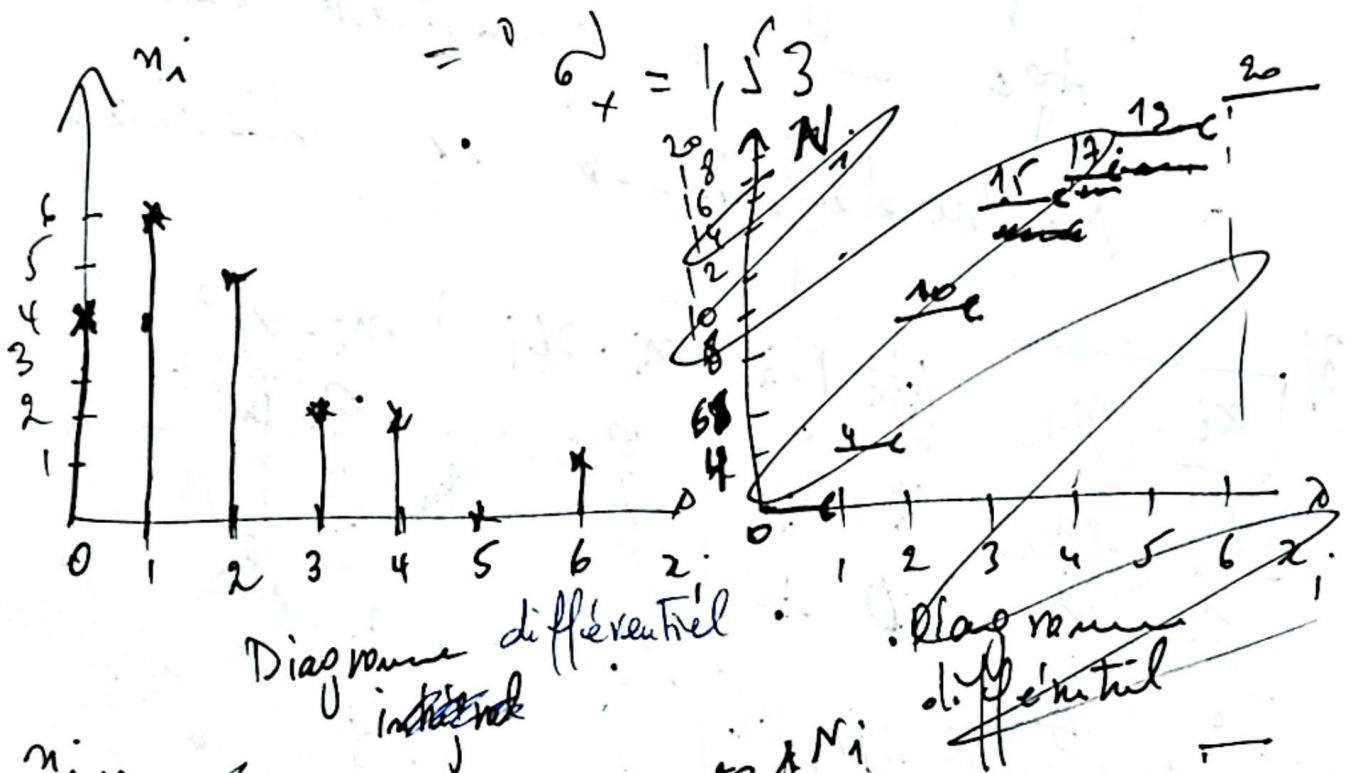
①

$$\bar{x} = \frac{\sum x_i \cdot n_i}{N} = \frac{36}{20} = 1,8 \triangleq 2$$

$$n_i = 1 < n_i < 2$$

$$s_x = \sqrt{v(x)}; v(x) = \frac{1}{N} \sum x_i^2 n_i - \bar{x}^2$$

$$= \frac{142}{20} - (1,8)^2 = 2,36$$



x_i	L_i	n_i	f_i	$C_i \cdot f_i$	N_i	F_i
64					0	0
68	66	15	0,075	4,95	15	0,075
72	70	33	0,165	11,55	48	0,24
76	74	26	0,13	9,62	74	0,37
80	78	52	0,26	20,28	126	0,63
84	82	32	0,16	13,12	158	0,79
88	86	30	0,15	12,15	188	0,94
92	90	12	0,06	5,14	200	1
Σ		200	1	77,82		

$$\bar{x} = \frac{\sum n_i x_i}{N} = 77,82$$

$$N' \in [76; 80]$$

$$N' = e_{i-1} + a_i \frac{0,5 - F_{i-1}}{f_i}$$

$$76 + 4 \left(\frac{0,5 - 0,37}{0,26} \right) = 78$$

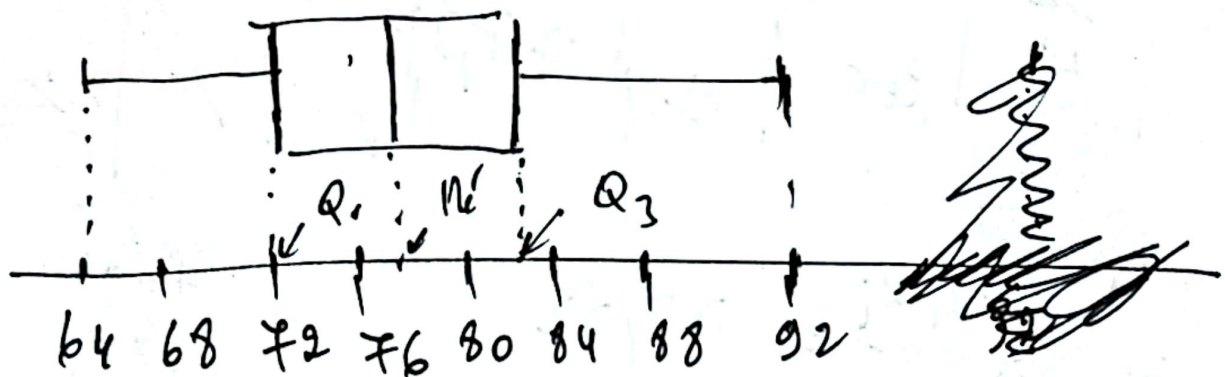
(3)

$$Q_2 = ? \quad Q_2 \in [72; 76]$$

$$Q_2 = 72 + 4 \frac{0,25 - 0,24}{0,13} = 72,3$$

$$Q_3 = e_{i-1} + a_i \frac{0,75 - F_{i-1}}{f_i}$$

$$= 80 + 4 \frac{0,75 - 0,63}{0,16} = 83$$



la médiane de l'échantillon est de poids entre
83 kg et 72,3 kg

2:03

x_i	y_i	x_i^2	$x_i \cdot y_i$	$\ln y_i$	$\ln y_i \cdot x_i$
1	4	1	4	1,386	1,386
2	12	4	24	2,484	4,969
3	35	9	105	3,555	10,666
4	109	16	436	4,681	18,765
5	320	25	1600	5,768	28,842
6	3	36	18	1,098	6,591
7	10	49	70	2,302	16,118
8	27	64	216	3,295	26,367
9	81	81	729	4,394	39,555
10	143	100	1430	4,962	49,627
Σ	55	744	385	4632	33,94
					202,87

$$1/ \bar{x} = \frac{1}{N} \sum x_i = \frac{55}{10} = 5,5$$

$$\bar{y} = \frac{1}{N} \sum y_i = \frac{744}{10} = 74,4$$

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2) Lo. variance de X

$$V(X) = \frac{1}{N} \sum_{i=1}^n x_i^2 - \bar{x}^2 = \frac{385}{10} - (5,5)^2 = 8,2$$

No. covariance

$$\begin{aligned} \text{cov}(X; Y) &= \frac{1}{N} \sum x_i y_i - \bar{x} \bar{y} \\ &= 463,2 - (5,5 \cdot 74,4) \\ &= 463,2 - 409,2 = 54 \end{aligned}$$

3) $Y = aX + b$ $\left\{ \begin{array}{l} a = \frac{\text{cov}(X; Y)}{V(X)} = \frac{54}{8,2} = 6,54 \\ b = \bar{Y} - a \bar{X} \\ = 74,4 - 6,545 \cdot 5,5 \end{array} \right.$

d'où $\boxed{Y = 38,4 + 6,545X}$

4) $\hat{Y}_{20} = 38,4 + 6,545(20)$
 $= 169$

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$$Y = B A^X \Rightarrow \ln Y = \ln B + X \ln A$$

~ pok: $Y' = B' + A' X$

~ $\text{cov}(X; Y') = \frac{1}{n} \sum X_i Y'_i - \bar{X} \bar{Y}' = 3$
 $20,298 - (5,5 \times 3,39) = 1,62$

$$\left\{ \begin{array}{l} A' = \frac{\text{cov}(X; Y')}{V(X)} = \frac{1,621}{8,27} = 0,196 \end{array} \right.$$

$$B' = \bar{Y}' - \bar{X} A' = 3,39 - 5,5 \times 0,196 = 2,31$$

$$B' = \ln B = p \quad B = e^{B'} = e^{2,31} = 10,1$$

$$A' = \ln A = p \quad A = e^{A'} = e^{0,196} = 1,2$$

$$Y = 10,1 \cdot 1,2^X$$

(7)